Arjun Kumar

Network Science Graduate Student

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Research Interests

- Voter behavior: network dynamics and influence maximization
- Sports analytics: influence, structures of markets and leagues
- International relations and economic policy: trade, capital, and influence flows
- Theoretical network science and graph theory

Education

Network Science Institute, Northeastern University | Boston, MA ..... 09/2026 – Present

M.S. in Complex Network Analysis

Graduate Certificate in Computational Social Science

University of North Dakota | Grand Forks, ND 05/2026

B.S. in Mathematics | **Capstone:** *Total Upsets from Seed:* NP-Complete Upset Cascades on College Football Season Networks
Major GPA: 3.82/4.0

Weber State University | Ogden, UT 04/2026

B.A in English, Creative Writing | Summa Cum Laude
Major GPA: 4.0/4.0

Minor in Economics

A.S. in Mathematics | High Honors 04/2025

Experience (Selected)

Network Science Data Fellow, AI I3 Accelerator | Boston Public Health Commission ..... 10/2026 – Present

- **Advisor:** Prof. Ajitesh Srivastava, Network Science Institute

Integrating heterogeneous public health data (EMS ambulance triage notes, resident 311 calls, syndromic surveillance encounter notes, and homicide-related news articles) directly within BPHC data systems, under the Commission's HIPAA, privacy, and data-dissemination policies. Working alongside BPHC staff and a faculty advisor on a pilot in the Commission's public-interest AI Insight-to-Impact Innovation accelerator, which connects Boston's academic partners with city public health priorities.

Python – NLP – SQL – Data Integration – Public Health Informatics – Quantitative Communication – Applied AI

Communication & Strategy Consultant | Adam Frisch for Congress, CO-3 ..... 08/2024 – 11/2024

Built models of fundraising and voter engagement from polling and demographic data; analyzed geographic and issue-level performance to guide district targeting and message testing.

Political Analytics – Fundraising – Voter Engagement – Demographic Analysis – Message Testing – District Targeting

Public Safety Officer (EMT) | Snowbird LLC ..... 11/2021 – 08/2026

Licensed and certified (lapsed) by Utah Bureau of Emergency Medical Services and National Registry of Emergency Medical Technicians. Responded to medical and security incidents in a high-risk mountain environment; documented incident data in coordination with law enforcement, dispatch, and hospital systems.

EMT Security Officer | Allied Security ..... 07/2021 – 11/2021

Responded to medical emergencies occurring among the employees of an advanced manufacturing facility, maintained accurate documentation of assessments and interventions, provided appropriate medical care.

Projects & Research

Revealed School Choice Markets in Chile | Network Science Project ..... *Ongoing*

- Constructed a national school-by-school co-application network from Chile's centralized admissions data (SAE) to evaluate the alignment between functionally revealed educational markets and administrative, political, and municipal boundaries (*comunas* & SLEP).

- Built a reproducible Python pipeline to process 1.61 million administrative choices from MINEDUC open data, extracting a bipartite applicant-school graph and projecting a national pre-K network of 4,640 institutions and 94,294 applicants.
- Adapted ϕ-correlation backbone extraction of Candia et al. (2019) and utilized community detection algorithms, investigating whether standard non-spatial filtering metrics artificially fragment the network.
- *Ongoing:* Currently benchmarking observed topology against randomized null models (controlling for institutional popularity and spatial costs) to investigate the role of socioeconomic assortativity where markets diverge from *comuna* or SLEP boundaries used in education funding and planning. Aiming for preprint and submission by Dec 2026.

Python – Network Science – Community Detection – Computational Social Science – Education Policy

Total Upsets from Seed | Capstone Research Project 12/2025

- **Advisor:** Prof. Timothy Prescott, UND
- Modelled NCAA football upsets as deterministic diffusion cascades on directed acyclic graphs encoding season structure; formulated the decision problem *Total Upsets from Seed (TUS)*: can a few early shocks trigger large-scale instability?
- Proved NP-completeness by reduction; ran Python simulations on toy graphs and a three-week slice of the 2025 Big 12 season to study how sparsity shapes cascades.
- Presented at the Undergraduate Research Showcase (Dec 2025).

Python – Network Science – Diffusion – Computational Complexity – Simulation – Sports Analytics

College Football Competitive Balance Study | Independent Data Analysis Project 05/2025

- Collected, cleaned, and analyzed multi-season team performance data from the Pac-12 using R. Applied key competitive balance metrics to create a formal report with data visualizations.

R – Data Visualization – Competitive Balance – Sports Analytics – Statistical Reporting

Talks & Publications

Solution to Problem 487 | Math Horizons (Mathematical Association of America) 02/2026
https://doi.org/10.1080/10724117.2025.2610148

"Boxer's Count" | American Mathematical Society Math & Poetry Contest 04/2025
Honorable Mention

"An Ecocritical Examination of Hemingway's Post-War Landscape" | National Undergraduate Literary Conference 03/2025
Paper Presentation, Advisor: Prof. Hal Crimmel, Weber State University

Expository Talks

"Resolution Limit in Community Detection" | presenting S. Fortunato & M. Barthélemy, 2007 ..... 10/2026

"School Choice as a Relational Urban Process" | presenting Y. Cortés, 2026 ..... 10/2026

"Optimal Sports League Realignment" | presenting N. Brixius, 2024 ..... 11/2025

Math Factor, Faculty Mentors: Profs. Cora Neal and Afshin Ghoreishi, Weber State University

Technical Skills

Languages: Python – R – SQL – LaTeX

Python Libraries: NetworkX – igraph – cdlib – graph-tool – infomap – python-louvain – pandas – NumPy – SciPy – scikit-learn – matplotlib – seaborn – statsmodels – tqdm – jupyter – sqlite3 – geopandas – OR-tools

R Packages: tidyverse – ggplot2 – dplyr – tidyr – purrr – tibble – stringr – forcats – fpp3 – forecast – tsibble – feasts – ggraph – igraph – tidygraph – broom – tidymodels

Network Software: Gephi – Cytoscape – NodeXL – Radatools

Other: Git/GitHub – Matlab – Mathematica – Stata – SPSS – Minitab – Excel

Certifications

Professional Certificate in Data Analytics with Excel and R | IBM 2025

Languages

English | Native

Spanish | Upper-intermediate working proficiency (B2)

Coursework

Quantitative

Northeastern University

CNET 5052 | Advanced Tools for Complex Network Analysis ..... Registered: SPR 2027
CNET 5314 | Complexity in Social Systems ..... Registered: SPR 2027
CNET 5442 | Sports Analytics Through Data and Networks ..... Registered: SPR 2027
CNET 5050 | Fundamentals of Complex Networks ..... In progress: FALL 2026
CNET 5051 | Analyzing Complex Network Data ..... In progress: FALL 2026
INSH 5301 | Introduction to Graduate Computational Statistics ..... In progress: FALL 2026

University of North Dakota

MATH 442 | Linear Algebra ..... SPR 2026
MATH 488 | Senior Capstone ..... FALL 2025
MATH 441 | Abstract Algebra ..... FALL 2025
MATH 266 | Differential Equations ..... SUM 2025
MATH 321 | Applied Statistical Methods ..... SPR 2025
MATH 208 | Discrete Mathematics ..... FALL 2024
STAT 500 | Computing for Statistics ..... FALL 202
ECON 411 | Economic Forecasting ..... SPR 2026

Weber State University

MATH 3610 | Graph Theory ..... FALL 2025
ECON 4810 | Sports Economics ..... SPR 2025
ECON 4010 | Intermediate Microeconomic Theory ..... SPR 2025

Thomas Edison State University

MAT 3510 | Mathematical Modeling ..... FALL 2025

Johns Hopkins University

110.301 | Introduction to Proofs ..... SPR 2025

University of Idaho

STAT 427 | R Programming ..... FALL 2024

Georgetown University

MATH 137 | Multivariable Calculus ..... SPR 2016
PECO 201 | Analytical Tools for Political Economy ..... FALL 2016
ECON 121 | Economic Statistics ..... FALL 2016
ECON 102 | Intermediate Macroeconomics ..... SPR 2016

Subject Matter Relevance

University of North Dakota

ECON 370 | Data Visualization ..... FALL 2025

Weber State University

ENGL 3100 | Technical Writing ..... SUM 2025
ECON 3200 | Money & Banking ..... SPR 2025

Georgetown University

GOVT 040 | Comparative Political Systems ..... FALL 2016
GOVT 060 | International Relations ..... SPR 2015
ECON 244 | International Finance ..... SPR 2015
ECON 243 | International Trade ..... FALL 2014
SOCI 196 | Comparative Law Enforcement ..... SPR 2016
PHIL 150 | Beginning Logic ..... SPR 2015
INAF 100 | States, Drugs, & Bugs of Africa ..... FALL 2014